

## Project Initialization and Planning Phase

|               |                                            |
|---------------|--------------------------------------------|
| Date          | 15 july 2026                               |
| Team ID       |                                            |
| Project Title | Voice Based Concept Understanding Analyser |
| Maximum Marks | 5 Marks                                    |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform concept learning and evaluation using **voice-based AI analysis**, improving understanding and feedback accuracy. It addresses the challenge of assessing conceptual clarity in spoken explanations, promising better learning outcomes, personalized feedback, and efficient evaluation. Key features include **speech-to-text conversion**, **semantic similarity scoring**, and **automated feedback generation**.

| Project Overview  |                                                                                                                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to enable learners to verbally explain concepts and receive instant feedback by comparing their spoken explanation with a reference answer using NLP and speech recognition.                  |
| Scope             | The project covers the entire process from audio input to feedback generation, including speech-to-text conversion, text preprocessing, semantic comparison, and report generation for personalized learning insights. |
| Problem Statement |                                                                                                                                                                                                                        |
| Description       | Students often struggle to evaluate their conceptual understanding when explaining topics verbally. Manual evaluation is time-consuming and subjective, leading to inconsistent feedback.                              |

|                          |                                                                                                                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Impact</b>            | Solving these issues will result in improved learning efficiency, objective evaluation, and enhanced confidence among learners, contributing to better educational outcomes.                                                              |
| <b>Proposed Solution</b> |                                                                                                                                                                                                                                           |
| <b>Approach</b>          | <b>Employing speech recognition and natural language processing (NLP) to analyze spoken explanations and compare them with reference answers, generating similarity scores and feedback reports.</b>                                      |
| <b>Key Features</b>      | - Speech-to-text conversion using Whisper or Azure Speech API. - Semantic similarity analysis using BERT or Sentence Transformers. - Real-time feedback and report generation. - Continuous improvement through adaptive learning models. |



## Resource Requirements

| Resource Type              | Description                             | Specification/Allocation |
|----------------------------|-----------------------------------------|--------------------------|
| <b>Hardware</b>            |                                         |                          |
| <b>Computing Resources</b> | CPU/GPU specifications, number of cores | T4 GPU                   |
| <b>Memory</b>              | RAM specifications                      | 8 GB                     |
| <b>Storage</b>             | Disk space for data, models, and logs   | 1 TB SSD                 |
| <b>Software</b>            |                                         |                          |
| <b>Frameworks</b>          | Python frameworks                       | Flask / FastAPI          |

|                                |                      |                                                                                 |
|--------------------------------|----------------------|---------------------------------------------------------------------------------|
| <b>Libraries</b>               | Additional libraries | Whisper, HuggingFace<br>Transformers, scikit-learn,<br>pandas, numpy            |
| <b>Development Environment</b> | IDE                  | Jupyter Notebook, PyCharm                                                       |
| <b>Data</b>                    |                      |                                                                                 |
| <b>Data</b>                    | Source, size, format | Custom dataset of concept<br>explanations and reference<br>answers (CSV format) |